

Wallace Peck

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PROFILE

Third Year Computer Science student with a specialisation in Artificial Intelligence, looking to leverage my technical expertise in Machine Learning to enhance Embodied AI platforms and foster Human-Robot collaboration. Fast learner and adept debugger. Strong work ethic and thrives in collaborative environments.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, C#, TypeScript
- **Frameworks:** PyTorch, ROS/ROS2, Webots, OpenCV, Docker, Git

EDUCATION

National University of Singapore (NUS) | Singapore **Aug 2023 - Apr 2027**

- Bachelor of Computing in Computer Science, specialisation in Artificial Intelligence **GPA: 4.27**
- **Relevant Courses:** Software Engineering (CI/CD), Foundations of Artificial Intelligence, Foundations of Machine Learning, Intelligent Robots: Algorithms and Systems

WORK EXPERIENCE

AGV Fleet Management System Engineer | Seagate Technology | Singapore **Jan 2026 - Jun 2026**

- Conducted research on industry and academic work on AMR deployments, algorithms
- Designed AMR simulator through requirements alignment, periodic feature reviews and iterative development
- Performed integration and system tests using replayed historical data to validate simulator correctness
- Trained nano ML model using PyTorch to enhance sensor data quality with low resource utilisation
- Proposed future improvements to support and enhance AMR deployments at Seagate sites

Full Stack Engineer | Lynk International | Singapore **Jul 2025 - Present**

- Harnessed AI coding tools to rapidly iterate SaaS website from prototype to production in 3 months
- Containerised deployment of back-end services on ECS; Serverless microservices for flexible loads
- Isolated Production, Testing and Development environments to increase tempo and reduce downtime
- Minimal cloud knowledge to secure and scalable deployment on AWS Cloud in one month

PROJECTS

Diffusion policy for Navigation | Personal Project **Oct 2025 - Dec 2025**

- Integrated ROS2 stack with Webots simulated home for experiments with General Navigation Models
- Trained alternate architecture using 400GB of combined datasets and compared with baseline models
- Explored action selection strategies to reduce path indecisiveness

Ultimate Tic-Tac-Toe Agent | National University of Singapore **March 2025 - April 2025**

- Sourced large external dataset to enhance model performance, trained on university compute cluster
- Scaled down existing model from 5M to 100K parameters (98% reduction) for competition constraints
- Attained 2nd place in intra-course competition, achieved parity with larger 5M model

Human and Objects Detection for Robotics Social Navigation | Ngee Ann Polytechnic **Apr 2020 - Apr 2021**

- Programmed a small, wheeled robot base with SR305 and ZED 2 vision sensors, powered by Nvidia Jetson Nano to create a social navigation agent
- Implemented obstacle avoidance algorithm from research paper for smooth and safe online navigation
- Designed a multi-language robotic software stack using ROS to integrate sensor-fusion, path planning and differential drive control